

We know that,

Link utilization of 100%,

$$U = \frac{t_f}{t_f + 2t_p}$$

when, window size is given then

$$U = \frac{w * t_f}{t_f + 2t_p}$$

Can be written as

$$U = \frac{w}{1 + 2 \frac{t_p}{t_f}}$$

$$\begin{aligned} U &= 100\% \\ &= \frac{100}{100} = 1 \end{aligned}$$

$$1 = \frac{w}{1 + 2A}, \text{ where } A = \frac{t_p}{t_f}$$

To achieve 100% utilization, window size

$$\underline{\underline{w = 1 + 2A}}$$

① A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs "Go-Back-N" with n set to 10. Assuming that each frame is 100 bytes long, what is the maximum data rate possible?

Sol:- Given,

$$\text{Data rate / Bandwidth} = 20 \text{ Kbps} \\ \Rightarrow R = 20 \times 10^3 \text{ bps}$$

$$\text{1 Frame length, } L = 100 \text{ bytes} = 100 \times 8 \\ = 800 \text{ bits}$$

$$\text{Propagation time, } \underline{t_p} = 400 \text{ ms}$$

$$\text{window size, } n = 10$$

$\Rightarrow$ Transmission time for 1 frame,

$$t_f = \frac{L}{R} = \frac{800}{20 \times 10^3} = 40 \times 10^{-3} \text{ sec} \\ = 40 \text{ msec}$$

$\Rightarrow$ Link utilization for 1 frame

$$= \frac{t_f}{t_f + 2t_p} = \frac{40}{40 + (2 \times 400)}$$

$$= \frac{40}{840} = 0.0476$$

$\Rightarrow$ Link utilization (in %) = 4.76%

$$\begin{aligned} \Rightarrow \text{For } n=10, \quad & 4.76 \times 10 \\ & = 47.6\% \\ & \approx 50\% \end{aligned}$$

So, Maximum data rate possible
when $B = 20 \text{ Kbps}$ is

$$20 \times \frac{50}{100} = \underline{\underline{10 \text{ Kbps}}}$$

② Consider 128×10^3 bps Satellite Communication link with one way Propagation delay of 150 msec.

Selective Retransmission (repeat) Protocol is used on this link to send data with a frame size of 1 KB. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____.

Sol:- Bandwidth, $R = 128 \times 10^3$ bps; $t_p = 150$ ms

Frame length, $L = 1 \text{ KB} = 1000 \text{ Bytes}$
 $= 8000 \text{ bits}$

Transmission time of frame, $t_f = \frac{8000}{128 \times 10^3}$ Seconds
 $t_f = 62.5 \text{ msec}$

Utilization, $U = 100\% = \frac{100}{100} = 1$

$$U = \frac{w}{1+2A} \Rightarrow 1 = \frac{w}{1+2A}$$

$$\Rightarrow w = 1+2A = 1 + 2 \left(\frac{t_p}{t_f} \right) = 1 + \frac{2 \times 150}{62.5}$$
$$w = 5.8 \approx 6$$

So, window size, w (or) no. of frames, $N \approx 6$

$$\text{Minimum no. of bits} = \lceil \log_2 2N \rceil$$

$$= \lceil \log_2 2 \times 6 \rceil$$

$$= \lceil \log_2 12 \rceil$$

$$= 4$$

So, minimum no. of bits required to represent the sequence number

$$= \underline{\underline{4}}$$

① Pure Aloha

$$\text{Efficiency } (\eta) = \underline{G * e^{-2G}}$$

where G = Number of stations willing to transmit data.

→ maximum efficiency occurs

$$\text{when } G = 1/2$$

$$\Rightarrow \eta = 1/2 * e^{-2 * 1/2}$$

$$= 0.184$$

Max. efficiency $\Rightarrow 18.4\%$

② slotted Aloha $\Rightarrow \underline{\eta = G * e^{-G}}$

→ maximum efficiency occurs

$$\text{when } G = 1$$

$$\Rightarrow \eta = 1 * e^{-1}$$

$$= 1/e$$

$$= 0.368$$

$$= 36.8\%$$

② A group of N stations share 100 Kbps Slotted ALOHA channel. Each station output a 500 bits frame on an average of 5000 ms even if previous one has not been sent. what is the required value of N ?

Sol:- $\rightarrow$ Throughput of one station:

$$= \text{No. of bits per Second}$$

$$= \frac{500 \text{ bits}}{5000 \text{ ms}}$$

$$= 500 / (5000 \times 10^{-3} \text{ s})$$

$$= 100 \text{ bps}$$

$\rightarrow$ Throughput of Slotted ALOHA

$$= \text{Efficiency} \times \text{Bandwidth}$$

$$= 0.368 \times 100 \text{ Kbps}$$

$$= 36.8 \text{ Kbps}$$

$\rightarrow$ Total No. of stations

$$\begin{array}{l} \text{Throughput of slotted} \\ \text{ALOHA} \end{array} = \begin{array}{l} \text{Total No. of} \\ \text{stations} \times \\ \text{Throughput at} \\ \text{each station} \end{array}$$

$$36.8 \text{ Kbps} = N * 100 \text{ bps}$$

$$N = \frac{36.8 * 10^3}{100}$$

$$= 36.8 * 10$$

$$\underline{\underline{N = 368}}$$

- Q A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at $2 * 10^8$ m/s. What is the minimum packet size that can be used on this network?

Sol:- Min. Packet size for a CSMA/CD LAN is the frame which covers whole RTT;

RTT = Round Trip Time

$$T_t = 2 * T_p$$

T_t = Transmission time

T_p = Propagation time

Distance, $d = 2 \text{ km} = 2 * 10^3 \text{ m}$

Velocity, $v = 2 * 10^8 \text{ m/s}$

Bandwidth, $B = 10^7 \text{ bps}$

$$T_p = \frac{D}{V}$$

$$= \frac{2 \times 10^3}{2 \times 10^8} \text{ Sec}$$

$$= 10^{-5} \text{ Sec}$$

L = length of frame

We know, $T_t = \frac{L}{B}$

$$= \frac{L}{10^7} \quad \text{--- (1)}$$

Also, $T_t = 2 * T_p$

$$= 2 * 10^{-5} \quad \text{--- (2)}$$

From (1) and (2)

$$\frac{L}{10^7} = 2 * 10^{-5}$$

$$\Rightarrow L = 2 * 10^{-5} * 10^7$$

$$= 2 * 10^2$$

$$L = 200 \text{ bits}$$

(or)

$$L = \frac{200}{8} = \underline{\underline{25}} \text{ bytes}$$